

Tips for Research

How to write a paper to be accepted



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<https://cvpr.kmu.ac.kr/KAIFRI/>



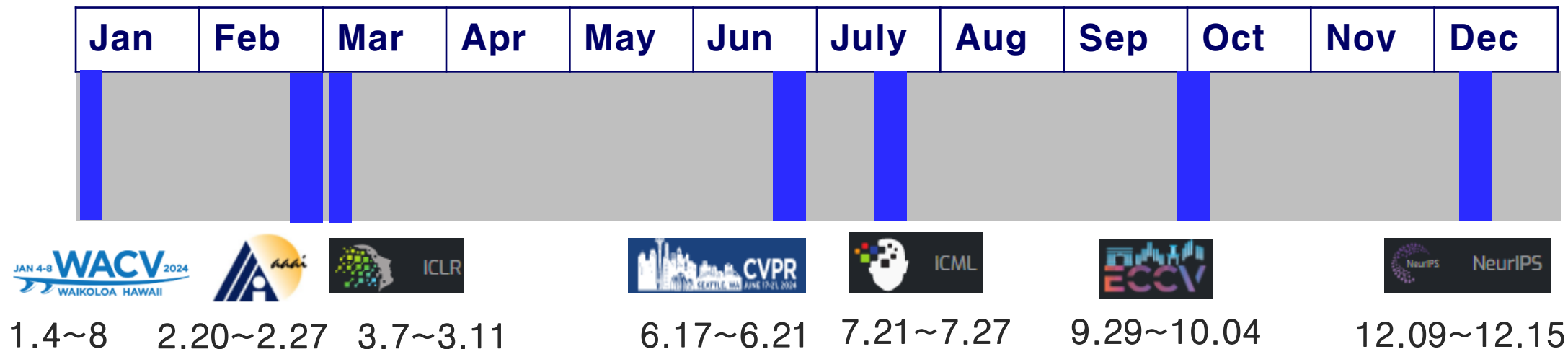
1. Feb. 04: Ph. D, Dept. Computer Science, Yonsei Univ.
2. Mar. 04 ~ Aug. 05: Senior Researcher, Samsung Elec.
3. Sept. 05 ~ Current: Professor, Dept. Com. Eng. KMU
4. Sept. 18 ~ Current: Vice Dean, Engineering College, KMU
5. Mar. 21 ~ Current: Chair, AI Research Center, KMU
6. Mar. 23 ~ Current: Chair, Daegu AI Researcher Forum
7. Jan. 24 ~ Current: Chair, AI&SP Society, IEIE
8. Selected Top 2% Scientist from Elsevier (Oct, 2023)
9. Received a Award from Ministry of Science and ICT (Feb. 2024)



Top Tier Conferences in AI, ML, CV

❖ 2024 AI Conferences

- The most important AI conference this year





❖ CVPR (IEEE/CVF Conference on Computer Vision and Pattern Recognition)

- Top Research Topics at CVPR
 - Artificial intelligence (94.95%)
 - Computer vision (42.90%)
 - Pattern recognition (32.71%)
- Focuses on Artificial intelligence, Computer vision, Pattern recognition, Feature extraction and Machine learning.

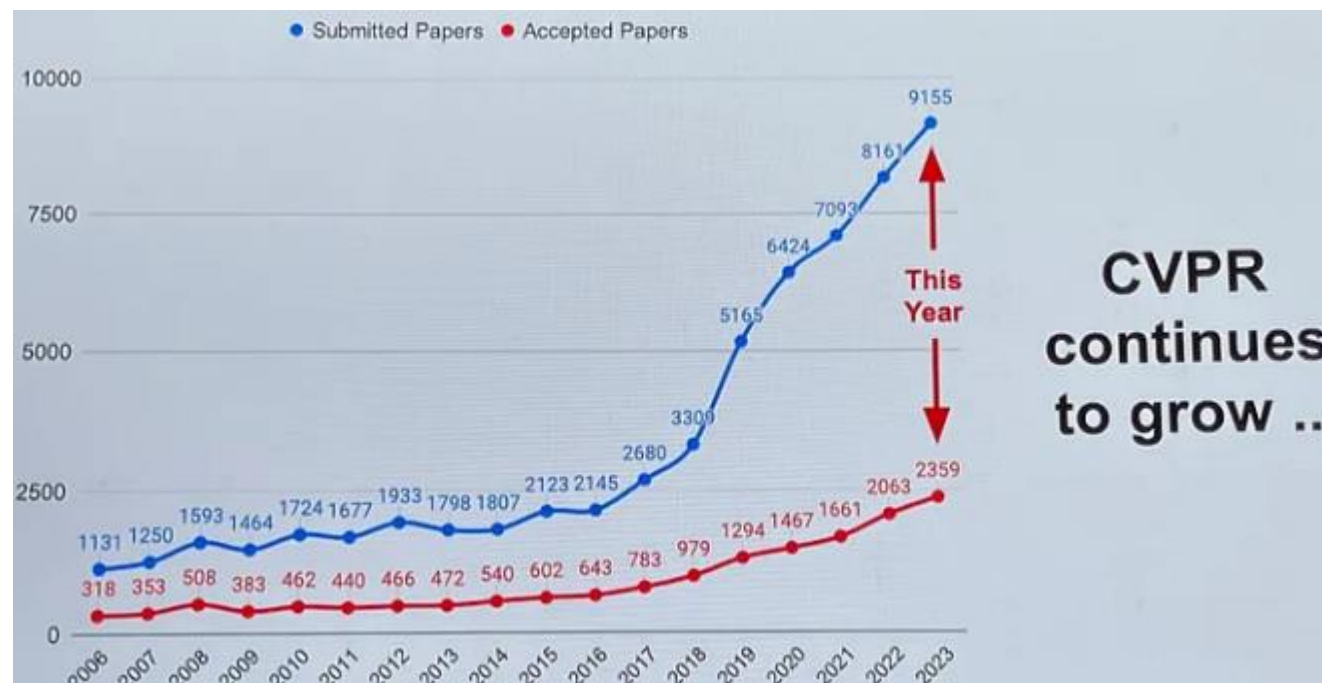


Top Tier Conferences in AI, ML, CV

❖ CVPR2023

■ Statistics CVPR2023

- Received 9155 submissions and accepted 2359 papers (25.8% acceptance rate)

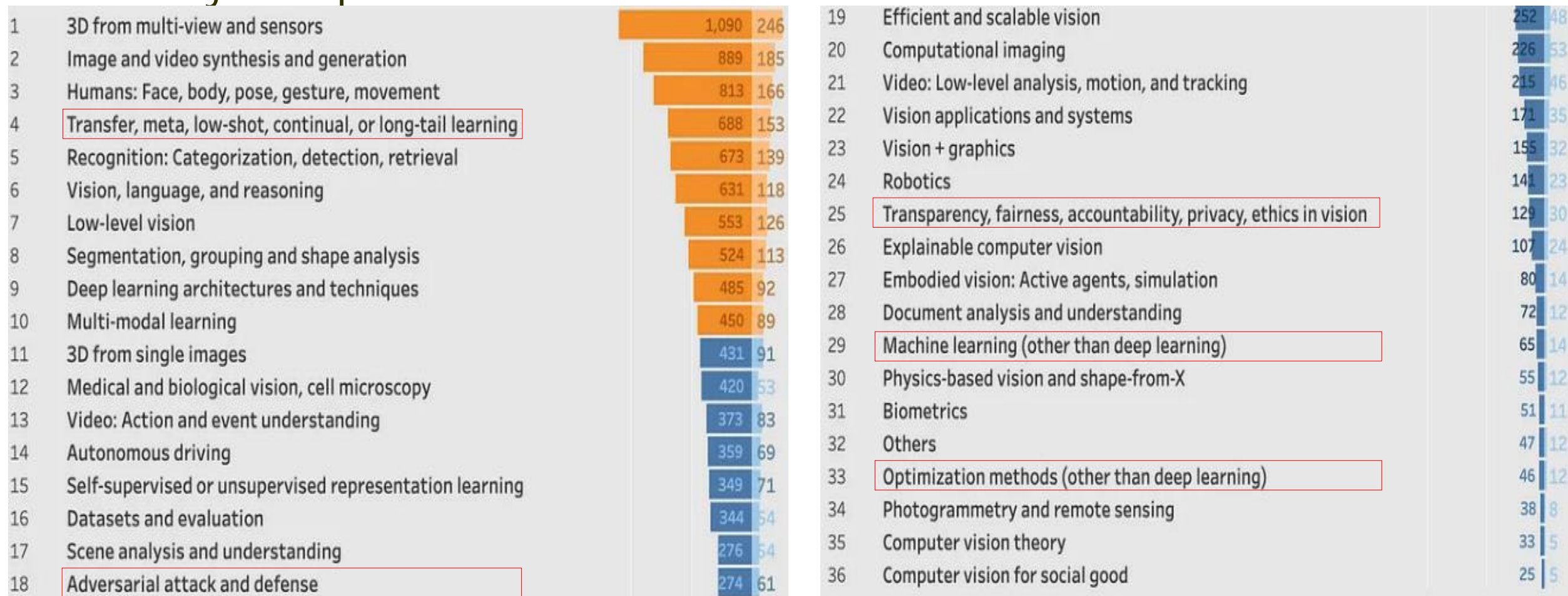




Top Tier Conferences in AI, ML, CV

❖ CVPR2023

■ Major Topics





❖ ICCV (IEEE/CVF International Conference on Computer Vision)

- Top Research Topics at ICCV
 - Artificial intelligence (94.84%)
 - Computer vision (43.04%)
 - Pattern recognition (29.60%)
- Focuses on Artificial intelligence, Computer vision, Pattern recognition, Feature extraction and Machine learning.

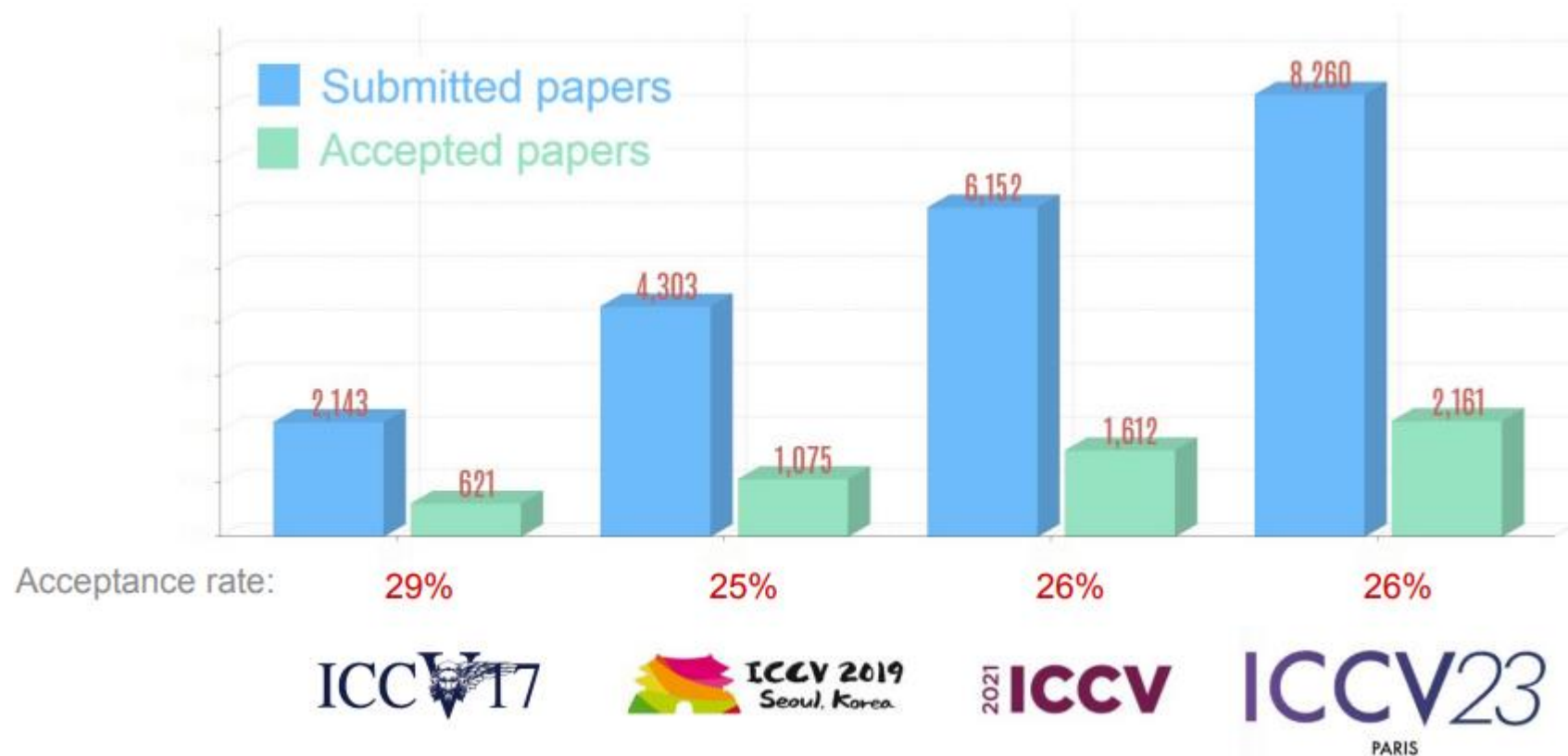


Top Tier Conferences in AI, ML, CV

❖ ICCV2023

■ Statistics ICCV2023

- Received 8,260 submissions and accepted 2,161 papers (26% acceptance rate)

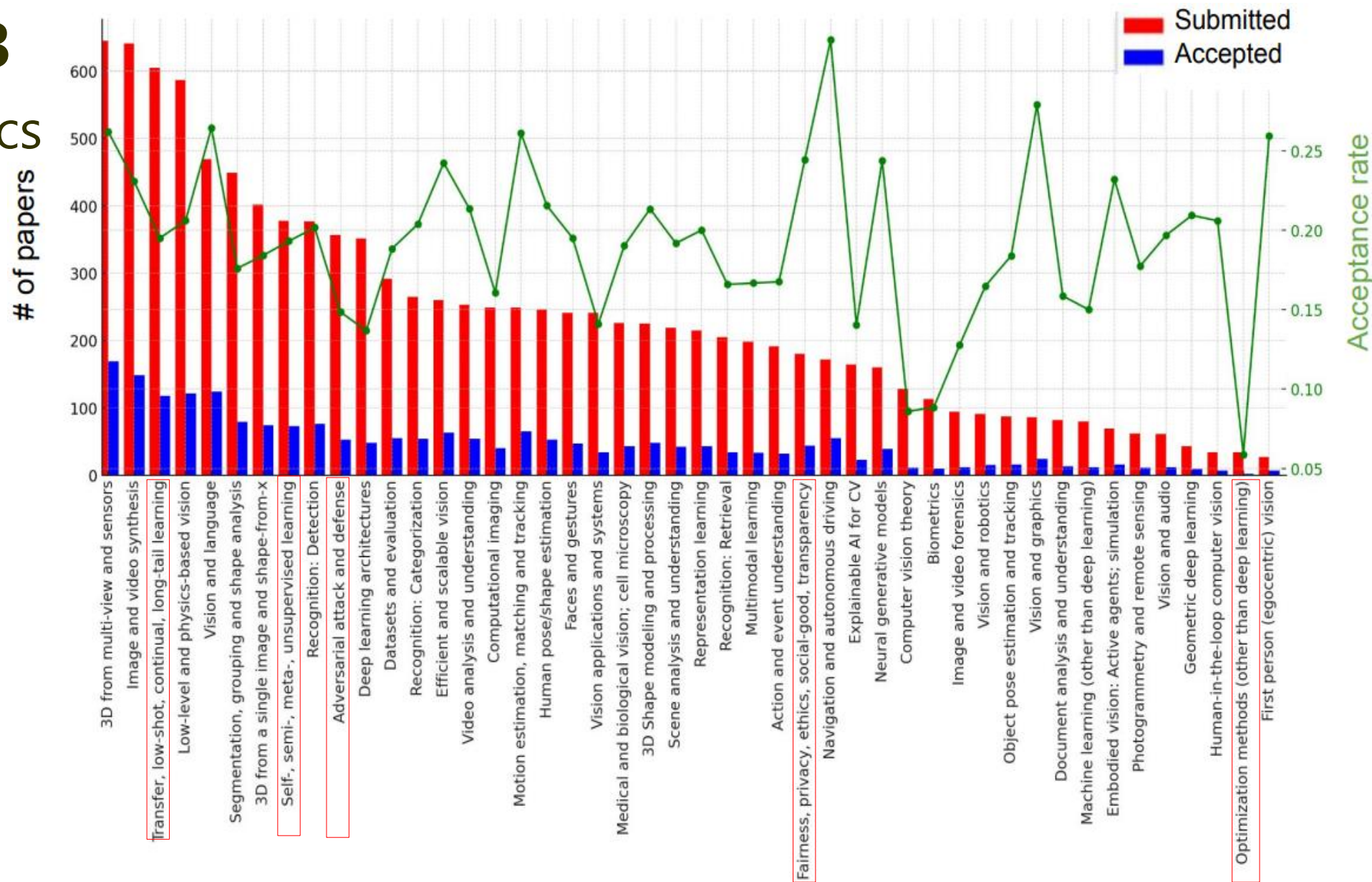




Top Tier Conferences in AI, ML, CV

❖ ICCV2023

■ Major Topics





❖ **ICML (International Conference on Machine Learning)**

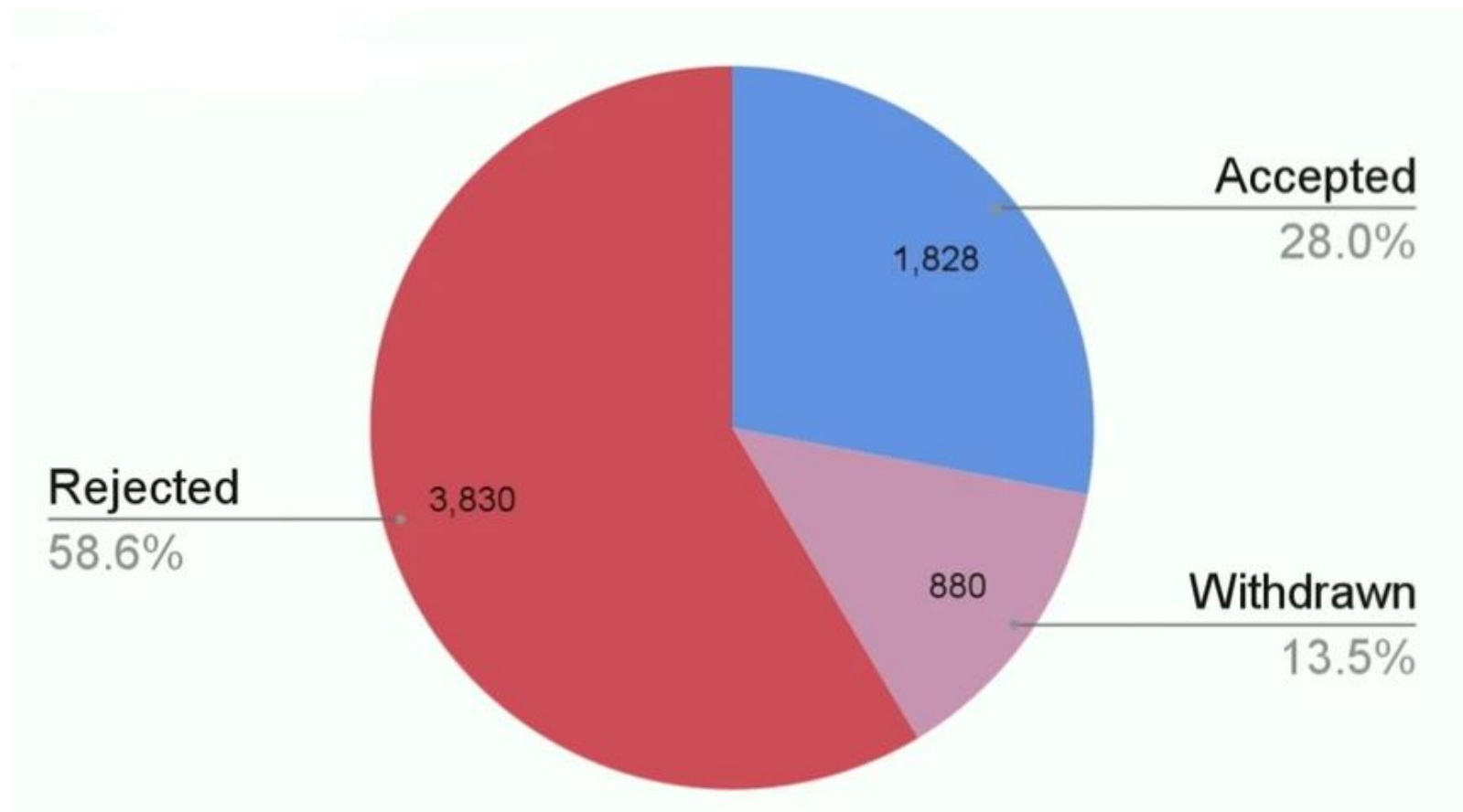
- Top Research Topics at ICML
 - Artificial intelligence (42.54%)
 - Machine Learning (22.37%)
 - Algorithm (15.66%)
- Focuses on Artificial intelligence, Machine learning, Algorithm, Mathematical optimization and Pattern recognition.



❖ ICML2023

■ Statistics ICML2023

- Received 6,538 submissions and accepted 1,828 papers (28% acceptance rate)





❖ ICML2023

■ Major Topics

- Machine Learning Fundamentals
- Deep Learning
- Computer Vision
- Natural Language Processing (NLP)
- Reinforcement Learning (RL)
- Explainable AI (XAI)
- Fairness, Accountability, and Transparency (FATE) in AI
- Applications of Machine Learning

many

few



❖ **AAAI (AAAI Conference on Artificial Intelligence)**

- Top Research Topics at AAAI
 - Artificial intelligence (42.0%)
 - Machine Learning (14.06%)
 - Theoretical computer science (9.03%)
 - ETC. (NLP, Mathematical optimization)
- Focuses on Artificial intelligence, Computer vision, Machine learning and Computer vision and Pattern recognition.



❖ AAI2024

- Statistics AAI2024
 - Received 9,590 submissions and accepted 2,340 papers (24.4% acceptance rate)

	#submissions	#accepts	Acceptance Rate
AAAI-24 (overall)	10,504	2,527	24.1%
Main Track	9,590	2,340	24.4%



❖ AAI2024

■ Major Topics

- Computer Vision
 - 3D Computer Vision > Language and Vision>Applications
- Machine Learning
 - Reinforcement Learning > Applications > GNN >Clustering
- Natural Language Processing
 - Applications> LLM
- Data Mining & Knowledge Management
- Human and AI
- ETC

many ↑

↓

few



❖ **NeurIPS (Annual Conference on Neural Information Processing Systems)**

- Top Research Topics at NeurIPS
 - Artificial intelligence (45.21%)
 - Machine Learning (20.21%)
 - Algorithm (19.19%)
- Focuses on Artificial intelligence, Machine learning, Algorithm, Mathematical optimization and Pattern recognition.



❖ **NeurIPS2023**

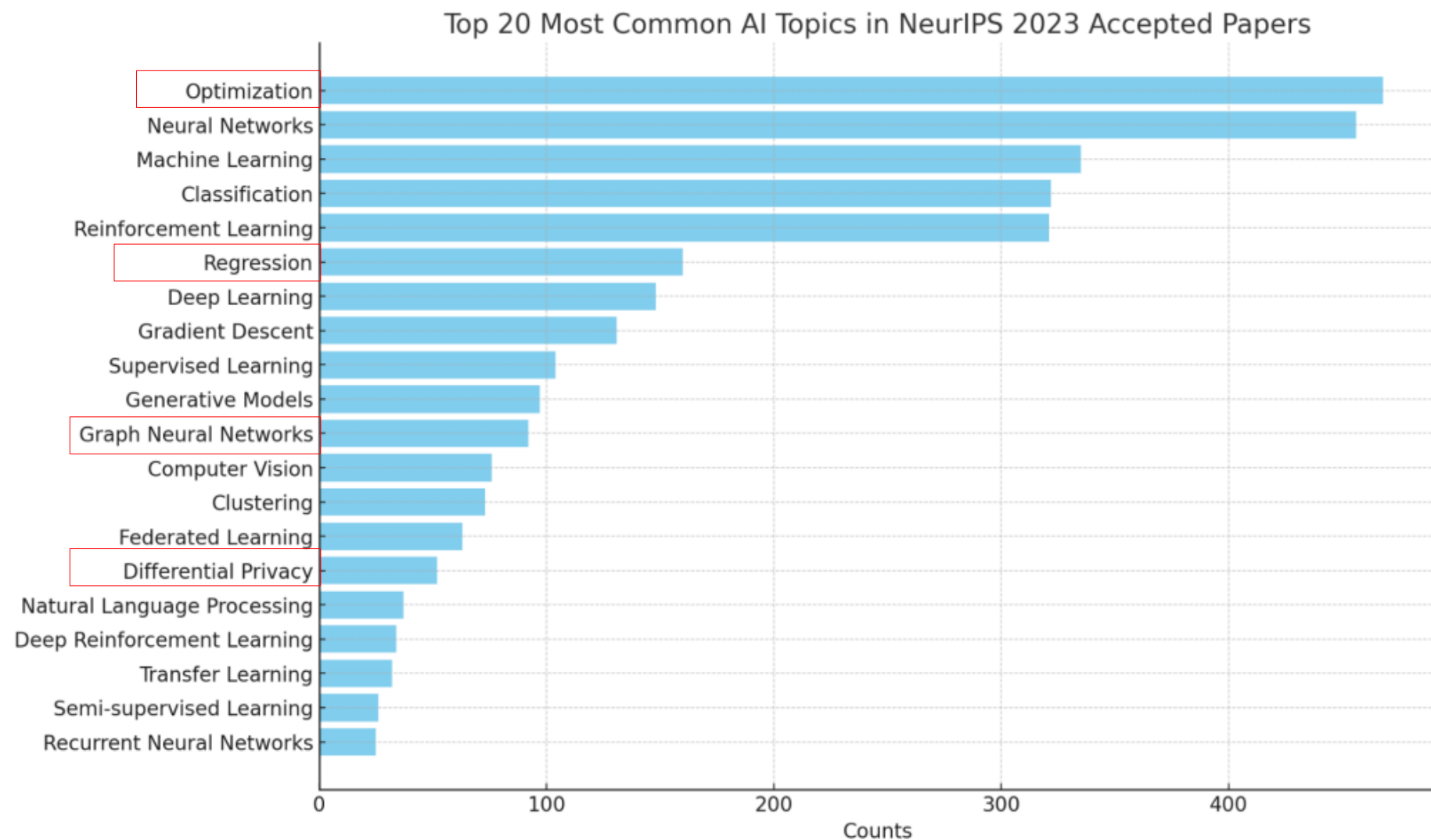
- Statistics NeurIPS2023
 - Received 12,343 submissions and accepted 3,218 papers (26.1% acceptance rate)



Top Tier Conferences in AI, ML, CV

❖ NeurIPS2023

■ Major Topics





❖ How papers are evaluated

- "After the papers come in"
 - Program chairs assign each paper to an area chair.
 - Area chairs assign each of their papers to 3~5 reviewers.
 - Reviewers read and review under 4~6 papers.
 - Authors respond to reviews (rebuttal).
 - Area chairs read reviews and author/reviewer discuss reviews & response / decide whether to reject or accept as poster or oral talk.
 - SACs & ACs make initial decisions (SACs focus on borderline cases)
 - ACs write metareviews (SACs focus on borderline cases)



❖ How papers are evaluated

- "After the papers come in"
 - PCs & SACs finalize decisions (ACs & reviewers involved as necessary)
 - Notification



❖ So, the Area Chair (AC) is the guy who decide your's destiny

- From an area chair's point of view, the types of papers in your pile
 - About 1/3 are obvious rejects
 - In the whole set, maybe 1 is a really nice paper-well-written, great results, good idea. **That will be an oral presentation.**
 - The rest are borderline, and these fall into two camps (reject or accept)...



Attitude when writing a paper

- ❖ **Need to pretend that you are the world's best experts in that field**
 - Need to know all relevant literature
 - Read them!
- ❖ **Need to persuade the reviewers that this method is very meaningful so need to be reported to other researchers in the field**
- ❖ **Need to be friendly to make reviewers understand your paper**
 - Every sentence needs to be crystal clear



Attitude when writing a paper

❖ Treat the reader as you would a guest in your house

- Anticipate their needs
 - "Would you like something to drink?"
 - "Something to eat?"
 - "Perhaps now, after eating, you'd like to rest?"





Attitude when writing a paper

- ❖ **No need to describe every details what you've done**
 - Paper is for letting people in community know the important discovery for advancing that field
 - It's not the detailed description of a small success in a limited benchmark
 - That's called "technical report" or "user's manual"
 - **You can even skip some details** if that does not well align with your story
 - "A paper is a logical nearest neighbor description of what you've done"
 - Larry S. Davis



Attitude when writing a paper

❖ Why the writing is important





❖ **Structure**

- Introduction
- Related work
- Approach (method)
- Experiments
- Discussion



❖ Jitendra Malik's advice

- A paper has four parts
 1. Title
 2. Abstract
 3. Introduction
 4. Rest of the paper

Spend equal time on all four of these!



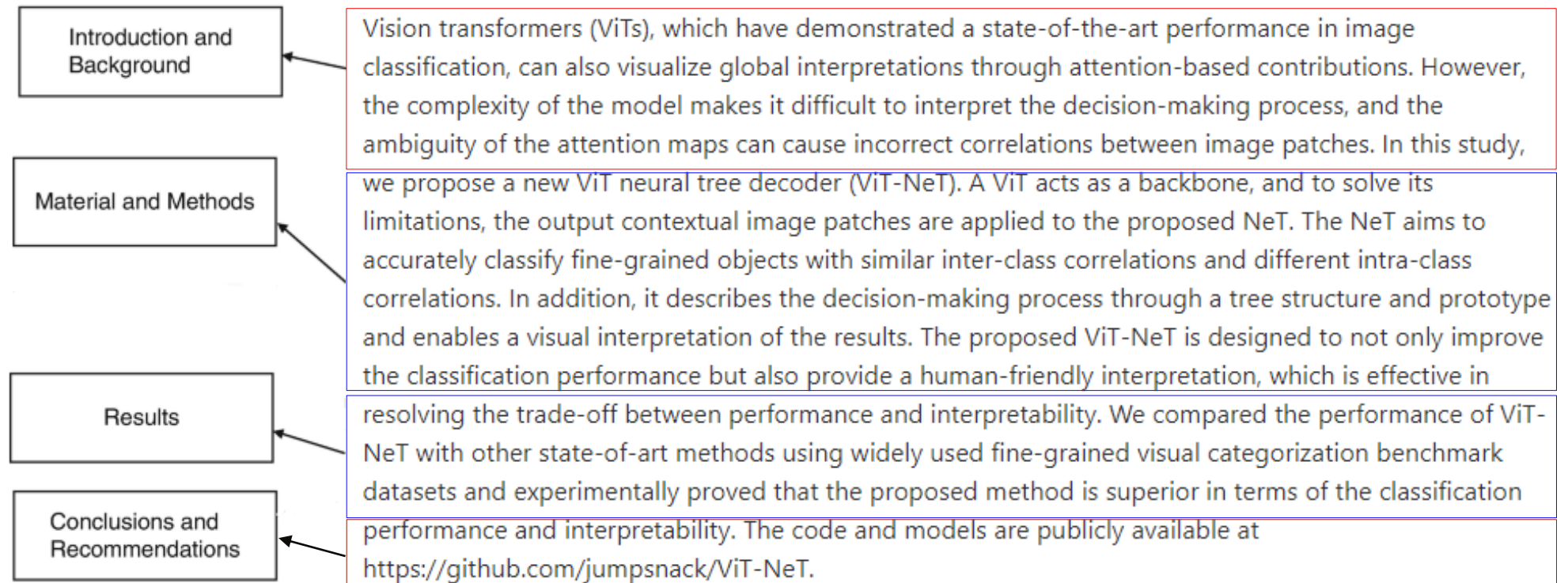
❖ **Tip 1: On the title**

- It should capture what is special about the paper.
- From the title reader should have a guess of the content of the paper, and recalling the title should help recall the paper.
- If every other paper in the field could use the same title, it is a bad title
 - "Deep learning based object detection" (bad)
 - "Attention is all you need" (Good), NeurIPS17, 116,693 cited
- Take the time to create terms that are easy to remember and specific.



❖ Tip 2: Abstract

- Give an overview of your entire research, in particular its findings and contribution to the field.
- Before reading the paper, readers can decide whether it is relevant to their research or not.





❖ Tip 3: Introduction section

- The most important section of a paper.
- For me (as a reviewer), once I have finished reading the introduction, I have formed an opinion of whether to accept or reject the paper
- What did you do (Fig. 1)

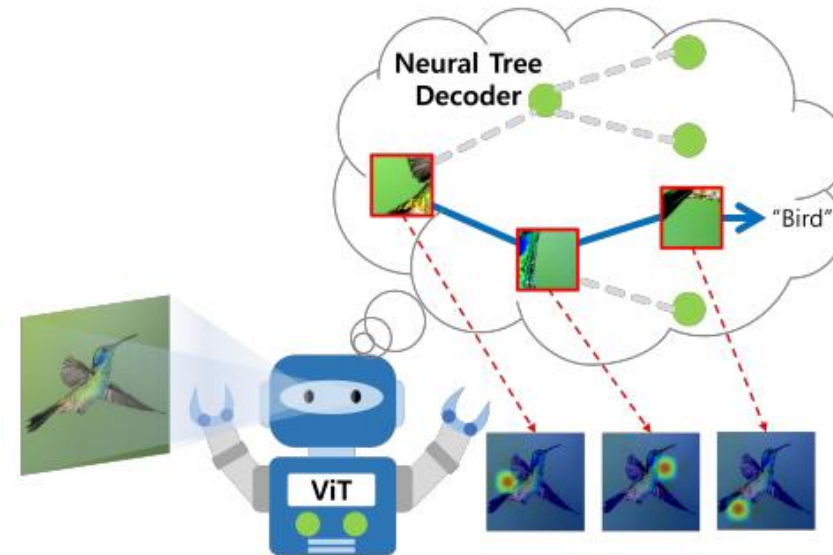


Figure 1. A ViT-NeT is a novel way of using a neural tree decoder to grant intrinsic interpretability to the ViT. In particular,



❖ Tip 3: Introduction section

- The most important section of a paper. For me, once I have finished reading the introduction, I have formed an opinion of whether to accept or reject the paper

CVPR2023
Best paper

Yihan Hu^{1,2*}, Jiazhi Yang^{1*}, Li Chen^{1*†}, Keyu Li^{1*}, Chonghao Sima¹, Xizhou Zhu^{3,1},
Siqi Chai², Senyao Du², Tianwei Lin², Wenhai Wang¹, Lewei Lu³, Xiaosong Jia¹
Qiang Liu², Jifeng Dai¹, Yu Qiao¹, Hongyang Li^{1†}

¹ OpenDriveLab and OpenGVLab, Shanghai AI Laboratory

² Wuhan University ³ SenseTime Research

*Equal contribution †Project lead

<https://github.com/OpenDriveLab/UniAD>

Abstract

Modern autonomous driving system is characterized as modular tasks in sequential order, i.e., perception, prediction, and planning. In order to perform a wide diversity of tasks and achieve advanced-level intelligence, contemporary approaches either deploy standalone models for individual tasks, or design a multi-task paradigm with separate heads. However, they might suffer from accumulative errors or deficient task coordination. Instead, we argue that a favorable framework should be devised and optimized in pursuit of the ultimate goal, i.e., planning of the self-driving car. Oriented at this, we revisit the key components within perception and prediction, and prioritize the tasks such that all these tasks contribute to planning. We introduce Unified Autonomous Driving (UniAD), a comprehensive framework up-to-date that incorporates full-stack driving tasks in one network. It is exquisitely devised to leverage advantages of each module, and provide complementary feature abstractions for agent interaction from a global perspective. Tasks are communicated with unified query interfaces to facili-

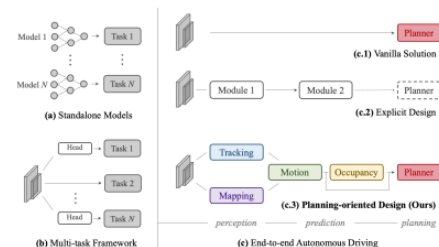


Figure 1. Comparison on the various designs of autonomous driving framework. (a) Most industrial solutions deploy separate models for different tasks. (b) The multi-task learning scheme shares a backbone with divided task heads. (c) The end-to-end paradigm unites modules in perception and prediction. Previous attempts either adopt a direct optimization on planning in (c.1) or devise the system with partial components in (c.2). Instead, we argue in (c.3) that a desirable system should be planning-oriented as well as properly organize preceding tasks to facilitate planning.

NeurIPS2023
Best paper

Privacy Auditing with One (1) Training Run

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Abstract

We propose a scheme for auditing differentially private machine learning systems with a single training run. This exploits the parallelism of being able to add or remove multiple training examples independently. We analyze this using the connection between differential privacy and statistical generalization, which avoids the cost of group privacy. Our auditing scheme requires minimal assumptions about the algorithm and can be applied in the black-box or white-box setting. We demonstrate the effectiveness of our framework by applying it to DP-SGD, where we can achieve meaningful empirical privacy lower bounds by training only *one* model. In contrast, standard methods would require training hundreds of models.

1 Introduction

Differential privacy (DP) [DMNS06] provides a quantifiable privacy guarantee by ensuring that no person's data significantly affects the probability of any outcome. Formally, a randomized algorithm M satisfies (ϵ, δ) -DP if, for any pair of inputs x, x' differing only by the addition or removal of one person's data and any measurable S , we have

$$\mathbb{P}[M(x) \in S] \leq e^\epsilon \cdot \mathbb{P}[M(x') \in S] + \delta. \quad (1)$$

A DP algorithm is accompanied by a mathematical proof giving an *upper bound* on the privacy parameters ϵ and δ . In contrast, a *privacy audit* provides an empirical *lower bound* on the privacy parameters. Privacy audits allow us to assess the tightness of the mathematical analysis [JUE20; NHSBTJCT23], or, if the lower and upper bounds are contradictory, to detect errors in the analysis or in the algorithm's implementation [DWWZK18; BGDCTV18; TTSSJC22].

Typically, privacy audits obtain a lower bound on the privacy parameters directly from the DP definition (1). That is, we construct a pair of inputs x, x' and a set of outcomes S and we estimate the probabilities $\mathbb{P}[M(x) \in S]$ and $\mathbb{P}[M(x') \in S]$. However, estimating these probabilities requires running M hundreds of times. This approach to privacy auditing is computationally expensive, which raises the question *Can we perform privacy auditing using a single run of the algorithm M ?*



❖ Tip 4: Write a dynamite introduction

- You must make your paper easy to read
- You've got to make it **easy** for anyone to tell what your paper is **about**, what problem it **solves**, why the problem is **interesting**, what is really **new** in your paper (and what isn't), why it's so **neat**.
- You must write a **dynamite introduction**.



❖ **Tip 5: Compare all relevant works**

- Even if it's the very first work in the your proposed set-up
 - Find the reasonable baselines
- Find out as many relevant work as possible
 - Make sure to include SOTA's (papers within 2 years)
 - You can exclude arxiv papers
 - If your method can't beat them
 - If your method can beat the super recent arxiv works
 - Why not include them! Compare and beat them!
- This way, you can avoid comment "weak experimental validation"



❖ Tip 5: Compare all relevant works

one module in the registration process, such as investigating more discriminate keypoint feature descriptors or more effective correspondence selection techniques, while the others [22, 30, 44] focus on registration in an end-to-end manner. However, deep-learned based methods require a large amount of data for training and usually lack generalization on different datasets. At present, it is still very challenging to achieve accurate registrations in the presence of heavy outliers and in cross-dataset conditions.

In this paper, we propose a geometric-only 3D registration method based on maximal cliques (MAC). The key insight is to loosen the previous maximum clique constraint, and mine more local information in a graph to generate accurate correspondences. We first model the initial correspondence graph, where each node represents a point cloud and each edge between two nodes represents a set of correspondences. Second, we search for the maximum clique in the graph and then use node-guided graph search to find each graph node with the appropriate correspondences. Compared with the maximum clique constraint and is able to mine more local information in a graph. This helps us to achieve plenty of correspondences from a graph. Finally, transformation is computed for the selected cliques by the SVD algorithm. The best hypothesis is selected to perform registration. Using popular hypothesis evaluation metrics in the RANSAC family. To summarize, our main contributions are as follows:

- We introduce a hypothesis generation method named MAC. Our MAC method is able to mine more local information in a graph, compared with the previous maximum clique constraint. We demonstrate that hypotheses generated by MAC are of high accuracy even in the presence of heavy outliers.
- Based on MAC, we present a novel PCR method, which achieves state-of-the-art performance on U3M, 3DMatch, 3DLoMatch and KITTI datasets. Notably, our geometric-only MAC method outperforms several state-of-the-art deep learning methods [3, 9, 19, 27]. MAC can also be inserted as a module into multiple deep-learned frameworks [1, 10, 18, 30, 44] to boost their performance. MAC combined with GeoTransformer achieves the state-of-the-art registration recall of 95.7% / 78.9% on 3DMatch / 3DLoMatch.

2. Related Work

2.1. Geometric-only PCR Methods

Various geometric-only methods [6, 8, 20, 37, 46] have been proposed recently. Typically, RANSAC and its variants [5, 13, 31, 32, 39–41] remain the dominant approaches

to the problem of estimating a 6-DoF pose from correspondences. RANSAC iteratively samples correspondences from the initial set, generating and evaluating geometric estimations for each subset until a satisfactory solution is obtained. Efficient and robust evaluation metrics are extremely important for using RANSAC to achieve accurate registration. To address the current problems of time-consuming and noise-sensitive evaluation metrics, [41] analyzes the contribution of inliers and outliers during the computation and proposed several metrics that can effectively improve the registration performance of RANSAC. A large number of variants have also been proposed to achieve further improvement. For example, Rusu *et al.* [32] presented the simple consensus-based initial alignment (SAC-IA) method, which samples correspondences spread out on the point cloud and leverages the Huber penalty for evaluation. Graph cut RANSAC (GC-RANSAC) [5] uses the graph-cut algorithm before model re-fitting in the local optimization step. Compatibility-guided sample consensus (CG-SAC) [31] additionally considers the normal information of key points during the sampling process. Yang *et al.* [40] proposed the sample consensus by sampling compatibility triangles (SAC-COT) method, which generates estimates by ranking and sampling ternary loops from the correspondence graph. Although many previous efforts have been made, these methods suffer from low time efficiency in cases with high outlier rates.

Some alternative methods based on BnB have been proposed. For example, Li *et al.* [42] proposed globally optimal methods which rationalizes the planning of local search stages, and its biggest advantage is to find the global optimum. Bustos and Chin [6] presented a novel method (GORE), which calculates the tight upper bound for each correspondence set by rejecting outliers. Motivated by GORE, Li [21] proposed a polynomial outlier removal method, which seeks the tight lower upper bound by calculating the costs of correspondence matrix (CM) and augmented correspondence matrix (ACM). However, BnB techniques are sensitive to the cardinality of the input and are time-consuming for large-scale inputs.

2.2. Deep-learned PCR Methods

In addition to geometric-only methods, recent works also adopt deep learning techniques to perform PCR. Some methods aim to detect more repeatable keypoints [4, 18] and extract more descriptive features [1, 10]. FCGF [10] computes the features in a single pass through a fully convolutional neural network without keypoint detection. D3Feat [4] uses a fully convolutional network to obtain local information of point clouds and a joint learning framework to achieve 3D local feature detection and description.

7 Related Work

Srivastava *et al.* [33] observed that while accuracy at a particular task can empirically appear sharp and unpredictable, cross-entropy does not appear so; the authors then discussed whether emergent abilities may be partially attributed to the metric. Our paper converts their discussion into precise predictions and quantitatively tests the predictions to reveal metric choice is possibly responsible for emergent abilities; well-known and widely-used metrics (including metrics used by previous works) have led to improvements; emergent abilities do not appear only on tasks involving multiple choice; discontinuous Multiple Choice Grade; metric choice can be used to induce emergent abilities in the domain (vision) in diverse architectures and tasks.

Alternative explanations for the origin of emergent abilities. Caballero *et al.* [5] explain emergence by assuming a power law functional form; under this view, emergent abilities are real, caused by a “break” (multiple breaks) in the governing power law. In contrast, our work suggests that emergent abilities are induced by the researcher under a single power law. Both

explanations could be true: some emergent abilities might genuinely be abruptly appearing, some emergent abilities might be attributable to the metric. Michaud *et al.* [28] posits that modeling data might be comprised of discrete subtasks (“quanta”) that networks learn; if networks have greater capacity, and are thus more capable of learning more of these quanta, then if some downstream task requires a network to learn some combination of quanta, larger networks are more likely to have all the requisite capabilities and thus are capable of performing this downstream task. We think that this is a very interesting hypothesis. Whether language modeling data can or should be understood from this quantization perspective, and whether these quanta indeed are the origin of emergent abilities, are really exciting questions that we think merit more study.

8 Discussion

Not over one column

Located before the conclusion

CVPR2023
Best paper

NeurIPS2023
Best paper



❖ Tip 6: Figure 1: WHAT did you do?

- Nothing is more frustrating than not being able to figure out what the paper is about until experiment part. 😡

- Show a TEASER figure on the very first page highlighting the inputs/outputs/key findings.

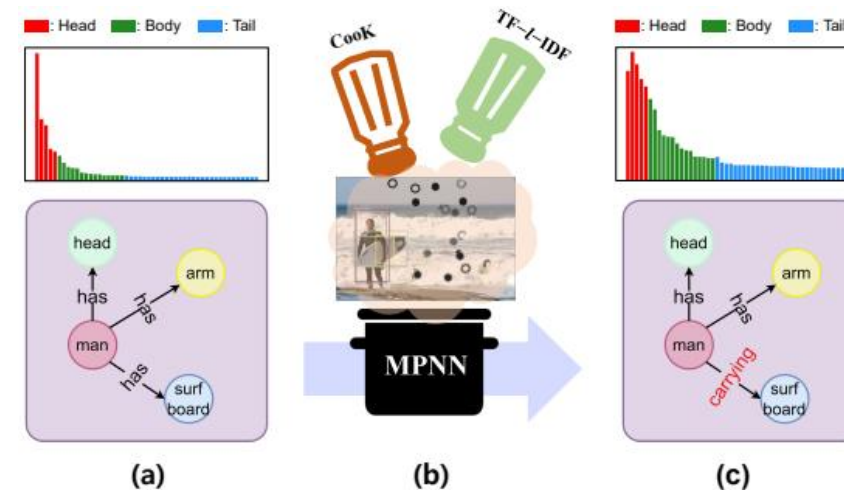
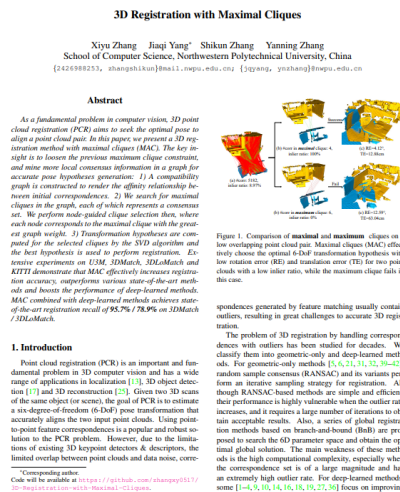


Figure 1. A novel learning recipe for SGG. (a) shows the class



❖ Tip 7: Figure 2: WHY did you do it?

- Motivate and justify the key insights/ideas of your work.
- Show an overview figure on how your method works.
- Label everything so that it provides a clear roadmap of the entire paper

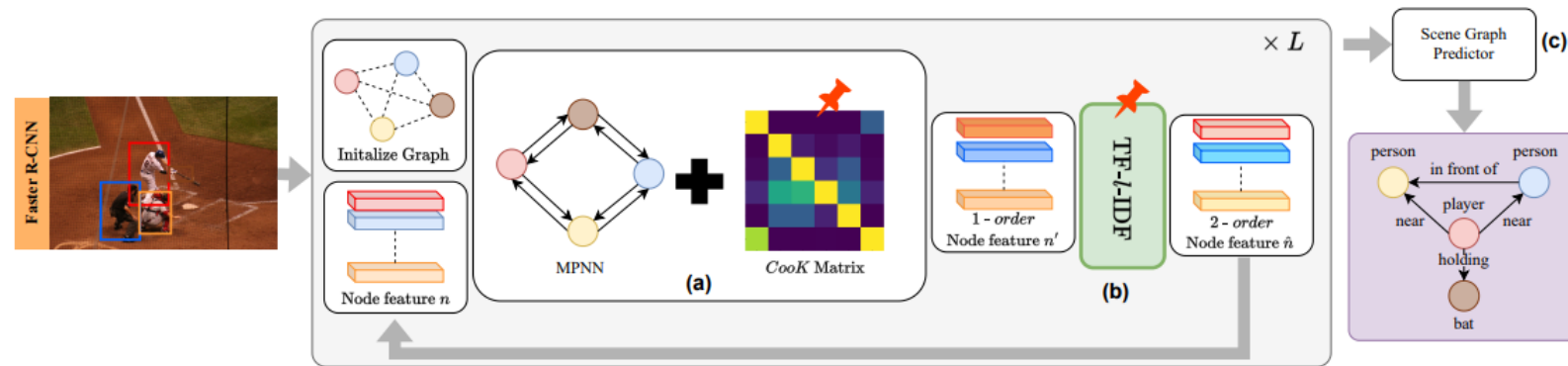


Figure 2. The whole training strategy of our proposed Cook + TF-l-IDF method. (a) In the MPNN process, we use the prior knowledge value $Cook(c_j|c_i)$ extracted from the training data to enable learning that reflects Cook. The 1-order node feature n' generated in this way is used as an input to (b) TF-l-IDF, which can update features by considering the frequency between labels, to create a 2-order node feature $\hat{n}$. Finally, the 2-order node feature $\hat{n}$ that has undergone L times of (a) and (b) processes is used to generate the final SG through the scene graph predictor in (c).



- ❖ **Tip 7: Figure 3~4: WHY did you do it?**
 - If you want to explain more concrete algorithm?
 - Detail structure of a core module
 - Do not over 4 figures for explaining algorithm
 - ICML, NeuralPS is not over 3

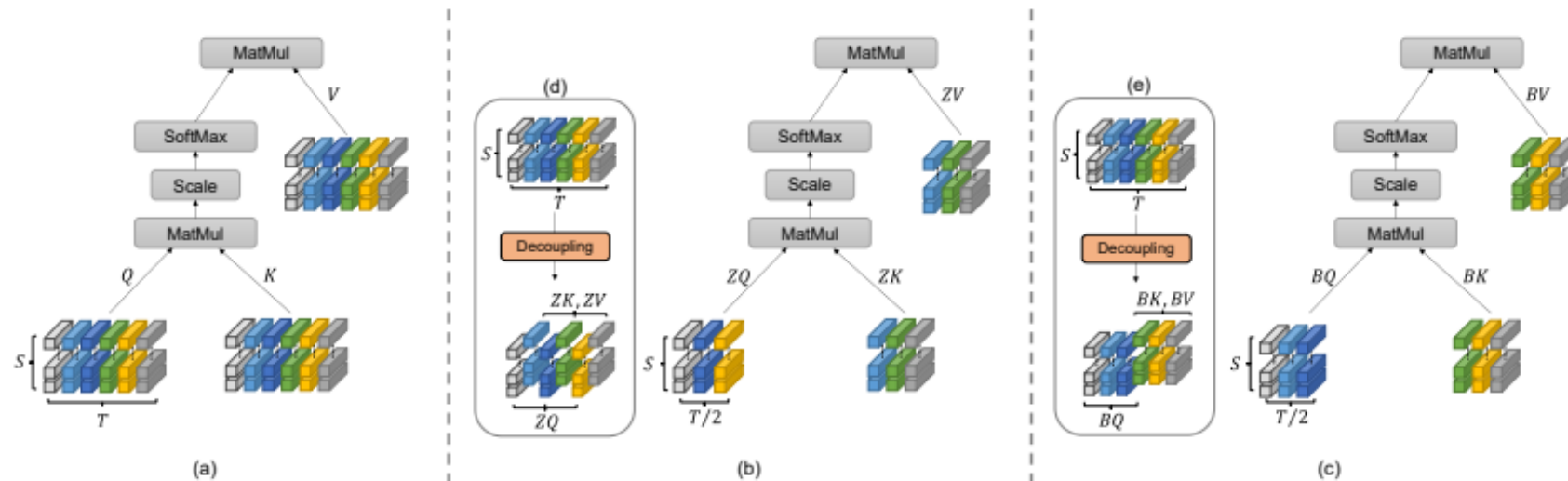


Figure 4. Proposed spatio-temporal cross attention modules. (a) full spatio-temporal attention (FAttn), (b) zigzag spatio-temporal attention (ZAttn), and (c) binary spatio-temporal attention (BAttn) modules.



❖ Tip 8: Self-contained figure/table caption

- Whatever you want to say for the figure/table, **say them in the caption.**
- It's annoying to find and match the corresponding texts describing the figure/table in your paper.
- Short titles in figure captions

Figures and captions

It should be easy to read the paper in a big hurry and still learn the main points. **Probably most of your readers will be skimming the paper.**

The figures and captions can help tell the story.

So the figure captions should be self-contained and **the caption should tell the reader what to notice about the figure.**

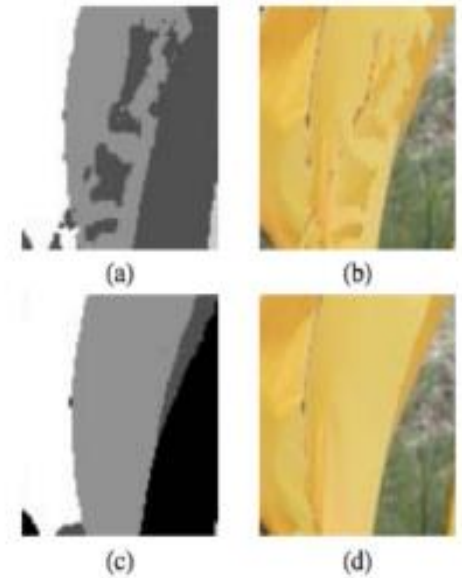


Figure 3: (a) Time-frame assignments for the front-most surface pixels, based on stereo depth measurements alone, without MRF processing. Grey level indicates the time-frame assignment at each pixel. (b) Shape-time image based on those assignments. (c) Most probable time-frame assignments, computed by MRF. (d) Resulting shape-time image. Note that the belief propagation in the MRF has removed spurious frame assignment changes.



❖ Tip 8: Concise notations and reduce number

- Use SINGLE letters for your math notations.
 - Examples:
 - $Color_j \Rightarrow C_j$
 - $Net \Rightarrow F(\cdot)$
- Use minimal number of equations
 - Too much details, makes your paper less understandable
 - Many good papers have few equations
 - AI, ML Conf. >> Vision Conf.



❖ Tip 9: Clean table

- Follow simple design principles for making clean a table:
 - no `\line`, use `\toprule`, `\midrule`, `\bottomrule`
 - no vertical lines
 - left align text
 - center align numbers
 - group and remove repetition with `multirow`/`multicol`

numeric literals	integers	in decimal	8743
		in octal	0o7464
			00103
		in hexadecimal	0x5A0FF
	fractionals		0xE0F2
		in decimal	140.58
			8.04e7
			0.347E+12
			5.47E-12
			47e22

Method	Pre-training	Feature			NTU60		NTU120	
		RGB	Est. Pose	Annot. Pose	XSub	XView	XSub	XSet
PoseMap [29]	✗	✓	✓		91.7	95.2	-	-
MMTM [35]	✗	✓		✓	91.9	-	-	-
VPN [14]	✗	✓		✓	95.5	98.0	86.3	87.8
DualHead-Net [10]	✗			✓	92.0	96.6	88.2	89.3
Skeletal GNN [51]	✗			✓	91.6	96.7	87.5	89.2
CTR-GCN [11]	✗			✓	92.4	96.8	88.9	90.6
InfoGCN [13]	✗			✓	93.0	97.1	89.8	91.2
3s-AimCLR [23]	✓			✓	86.9	92.8	80.1	80.9
PoseC3D [17]	✓	✓	✓		97.0	99.6	95.3	96.4
KA-AGTN [31]	✗			✓	90.4	96.1	86.1	88.0
STAR-Transformer	✗	✓		✓	92.0	96.5	90.3	92.7

Table 2. Performance comparison with other state-of-the-art methods on the NTU-RGB+D Action dataset (Est, estimated pose(skeleton); XSub, cross-subject; XView, cross-view; XSet, cross-setup).

How to write a paper



❖ Tip 10: Avoid empty spaces

- Fill the paper into full page limit.
- It gives your readers a sense of a POLISHED and not RUSHED paper.

Scene Graph Generation Strategy with Co-occurrence Knowledge and Learnable Term Frequency



Figure 3. Difference in TF-*l*-IDF performance according to the batch size. As the proposed TF-*l*-IDF is performed in batches, it can be confirmed that the performance increases proportionally as the batch size increases.

Cook-Type	SGGGen		
	mR@20	mR@50	mR@100
One Cook for VG	11.1	14.2	16.3
Advanced	11.4	14.9	17.1

Table 5. Performance changes when using Advanced Cook, a prior knowledge collected from a wider variety of environments. Advanced Cook achieved a higher performance than when it was not used. Because the OI dataset has more object labels than the VG dataset, $Map : OI \rightarrow VG$ was excluded as many cases occurred where mapping was not possible.

to perform more sophisticated feature updates.

4.6. Long-tail Alleviation

In this experiment, we analyzed the effects of the proposed learning method on long-tail problem mitigation. Figure 4 shows the change in mR@100 for each class when the proposed method was applied. As illustrated in the figure, the mR@100 value for the head decreased, whereas those for the body and tail parts increased significantly. This demonstrates that Cook’s ‘knowledge of object co-occurrence’ and TF-*l*-IDF’s ‘feature update’ were successfully applied to each class part.

4.7. Advanced Cook

Human knowledge has become more generalized and reliable owing to extensive experience and activities. To verify the applicability of this human knowledge paradigm to Cook, we generated an advanced Cook based on additional datasets and applied it to the model. In this experiment, we combined two Cooks from the VG and OI datasets to create an advanced Cook. The mapping function $Map : OI \rightarrow VG$ for the combination was hand-crafted. Table 5 shows the performance results when using advanced Cook. Similar to the general improvement effect of knowledge, the advanced Cook achieved a higher performance than individual Cook. This demonstrates that Cook can im-

prove the performance if the task uses knowledge obtained from similar datasets.

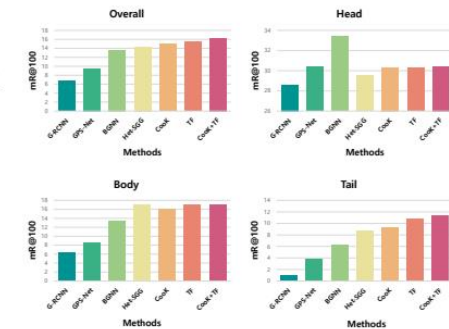


Figure 4. TF-*l*-IDF effect on long-tail problem. The proposed TF-*l*-IDF successfully reduces the mR@100 for common labels in head and focuses more on rare labels in body and tail.

5. Conclusion

In this study, we proposed a Cook and TF-*l*-IDF layer that can solve co-occurrence knowledge and long-tail problems. Our proposed method has a significant advantage in that it can improve the performance of SGG tasks because it can be easily applied without significantly changing the existing model. In addition, by performing experiments using advanced Cook, we verify that this study realized a new approach for generating more general knowledge matrices. However, limitation is that it can only be applied to existing MPNN-based models. In addition, Cook learning is currently applicable only to supervised learning; therefore, it is difficult to apply it to foundation models conducted with self-supervision. In future research, we plan to study SGG, which enables Cook to learn the relationships between objects on its own and is applicable regardless of the SGG model.



❖ Tip 11: Use concise description

- Do not be verbose!

Re-writing exercise

The underlying assumption of this work is that the estimate of a given node will only depend on nodes within a patch: this is a locality assumption imposed at the patch-level. This assumption can be justified in case of skin images since a pixel in one corner of the image is likely to have small effect on a different pixel far away from itself. Therefore, we can crop the image into smaller windows, as shown in Figure 5, and compute the inverse J matrix of the cropped window. Since the cropped window is much smaller than the input image, the inversion of J matrix is computationally cheaper. Since we are inferring on blocks of image patches (i.e. ignoring pixels outside of the cropped window), the interpolated image will have blocky artifacts. Therefore, only part of xMAP is used to interpolate the image, as shown in Figure 5.

Before

We assume local influence--that nodes only depend on other nodes within a patch. This condition often holds for skin images, which have few long edges or structures. We crop the image into small windows, as shown in Fig. 5, and compute the inverse J matrix of each small window. This is much faster than computing the inverse J matrix for the input image. To avoid artifacts from the block processing, only the center region of xMAP is used in the final image, as shown in Fig. 5.

After

This editing benefits you twice: (1) you have 50% more space to tell your story, and (2) the text is easier for the reader to understand.



❖ **Tip 12: Tone: be kind and gracious**

- Do not use decisive tone if you are not 200% sure
 - Instead, use defensive tone. For example,
 - We believe...
 - We conjecture...
- Write in inclusive manner.
 - Make readers (reviewers) happy
 - They are likely the authors of your competing methods



❖ **Tip 13: Develop a reputation for being clear and reliable**

- Don't overdo your research, hide your shortcomings, and disparage other people's work (it's not in your long-term and short-term interests).
 - Example) "because the author was Albert, I knew I could trust the results." [a conference chair discussing some of the reasons behind a best paper prize selection]



- ❖ **Tip 14: Do not ever overstate or overpromise**
 - Always support your arguments by experiments or proof
 - Otherwise, do not say it!



❖ Tip 15: Use supplementary material

- Add more results and large figures
- Add technical details as necessary (don't miss important details)
- Derivation details,
 - e.g., proof of a theorem

Cross-Modal Learning with 3D Deformable Attention for Action Recognition

Supplimentary Materials

1. Detailed Description of Stride Attentions

In general, most of the time complexity of transformers is highly related to the attention operation. We offer stride attentions for efficient correlation learning between discretized tokens. In this section, we describe details of the proposed joint stride attention and temporal stride attention.

1.1. Joint Stride Attention

The number of *pose* tokens (P) can grow dynamically based on the number of joints (R) and people appeared in a scene. Since this is directly related to the amount of computation required for attention, we propose joint stride attention dividing *pose* tokens into sliding windows. Algorithm 1 describes detailed operation of the proposed joint stride attention. According to the notations used in the Algorithm 1, we can organize comparisons of time complexity between full attention and joint stride attention as described in Table 1.

Table 1: Complexity comparisons between full attention and joint stride attention

Method	Complexity
Full attention	$O(T^2 R^2)$
Joint stride attention	$O(T^2 wnd^2)$

In joint stride attention, we decompose *pose* tokens using sliding window (wnd) with a *stride* having halved size of wnd . If *pose* tokens $P \in \mathbb{R}^{4C \times T \times R}$ are fed to full attention, the time complexity per layer becomes $O(T^2 R^2)$ and the equation is as follows:

$$\mathbf{O} = [P || M_{pose} || M'_{CLS}] \quad (1)$$

$$\text{FullAttention}(\mathbf{O}\mathbf{W}_q, \mathbf{O}\mathbf{W}_k, \mathbf{O}\mathbf{W}_v) = \sum_t^T \sum_r^R \text{softmax}\left(\frac{\mathbf{O}\mathbf{W}_q^{t,r} \mathbf{O}\mathbf{W}_k^{t,r}}{\sqrt{d_h}}\right) \mathbf{O}\mathbf{W}_v^{t,r} \quad (2)$$

However, when *pose* tokens are fed to joint stride attention, the time complexity per layer becomes $O(T^2 wnd^2)$ and the equation is as follows:

$$\text{JointStrideAttention}(\mathbf{O}\mathbf{W}_q, \tilde{\mathbf{O}}\mathbf{W}_k, \tilde{\mathbf{O}}\mathbf{W}_v) = \sum_t^T \sum_w^{wnd} \text{softmax}\left(\frac{\mathbf{O}\mathbf{W}_q^{t,w} \tilde{\mathbf{O}}\mathbf{W}_k^{t,w}}{\sqrt{d_h}}\right) \tilde{\mathbf{O}}\mathbf{W}_v^{t,w} \quad (3)$$

where $\mathbf{O}$ and $\tilde{\mathbf{O}}$ from Algorithm 1. In Eq. 3, since wnd is always less than R , the overall time complexity of joint stride attention is smaller than full attention.

1.2. Temporal Stride Attention

Temporal stride attention is proposed to capture temporal changes between each sequential frame and joint. The overall procedure is described in Algorithm 2. The number of input tokens D_n is sum of the number of *RGB*, *pose* and cross modal tokens. To decompose these tokens into small temporal windows, we apply sliding window along with the temporal dimension. The complexity comparison result between full attention and temporal stride attention is described in Table 2.

Table 2: Complexity comparisons between full attention and temporal stride attention

Method	Complexity
Full attention	$O(T^2 D_n^2)$
Temporal stride attention	$O(wnd^2 D_n^2)$

In the case of full attention, the time complexity of attention against the concatenated tokens $\mathbf{N} \in \mathbb{R}^{4C \times T \times D_n}$ becomes $O(T^2 D_n^2)$ and the equation is as follows:

$$\mathbf{N} = [Z || P || M_{RGB} || M_{pose} || M_{CLS}] \quad (4)$$

$$\text{FullAttention}(\mathbf{N}\mathbf{W}_q, \mathbf{N}\mathbf{W}_k, \mathbf{N}\mathbf{W}_v) = \sum_t^T \sum_n^{D_n} \text{softmax}\left(\frac{\mathbf{N}\mathbf{W}_q^{t,n} \mathbf{N}\mathbf{W}_k^{t,n}}{\sqrt{d_h}}\right) \mathbf{N}\mathbf{W}_v^{t,n} \quad (5)$$

On the contrary, when the concatenated tokens are decomposed into several sliding windows, the time complexity per layer becomes $O(wnd^2 D_n^2)$ and the equation is as follows:



❖ Tip 15: Use supplementary material

- Add more results and large figures
- Add technical details as necessary (don't miss important details)
- Derivation details,
 - e.g., proof of a theorem

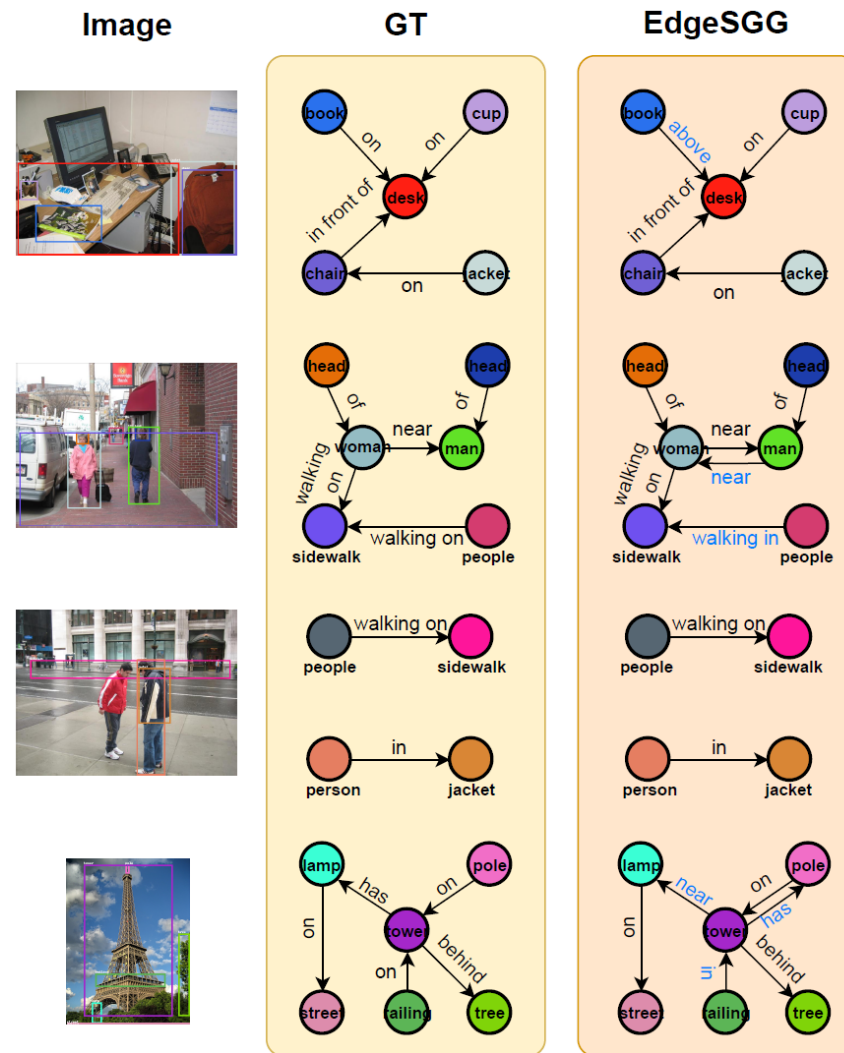


Figure A1: Additional visualization results using EdgeSGG on PredCls task. As depicted in the figure, the model demonstrates improved accuracy in fine-grained relations.



❖ Tip 15: Use supplementary material

- Add more results and large figures
- Add technical details as necessary (don't miss important details)
- Derivation details,
 - e.g., proof of a theorem

Algorithm 1 Joint stride attention

Input: *Pose* tokens P , Memorized *CLS* modal token M_{CLS} , *Pose* modal token M_{pose} , window size wnd , query weight W_q , key weight W_k , value weight W_v

$stride \leftarrow \lfloor wnd/2 \rfloor$

$\hat{P}_q \leftarrow \emptyset$ ▷ query set

$\hat{P}_{kv} \leftarrow \emptyset$ ▷ key and value set

$i \leftarrow 0$

while $i < (R - wnd)$ **do** ▷ Split P into query set $\hat{P}_q$

$\hat{P} \leftarrow P[:, i : i + wnd]$ ▷ $P \in \mathbb{R}^{4C \times T \times R}$

$\hat{P}_q \leftarrow \hat{P}_q \cup \hat{P}$ ▷ $\hat{P} \in \mathbb{R}^{4C \times T \times wnd}$

$i \leftarrow i + stride$

if $i > (R - wnd)$ **then**

$i \leftarrow i - wnd$ ▷ To assure query covers entire tokens

end if

end while

$i \leftarrow stride$

while $i < (R - wnd)$ **do**

▷ Split P into key and value set $\hat{P}_{kv}$

$\hat{P} \leftarrow P[:, i : i + wnd]$ ▷ $P \in \mathbb{R}^{4C \times T \times R}$

$\hat{P}_{kv} \leftarrow \hat{P}_{kv} \cup \hat{P}$ ▷ $\hat{P} \in \mathbb{R}^{4C \times T \times wnd}$

$i \leftarrow i + stride$

end while

$D_q \leftarrow |\hat{P}_q|$ ▷ $\hat{P}_q \in \mathbb{R}^{D_q \times 4C \times T \times wnd}$

$D_{kv} \leftarrow |\hat{P}_{kv}|$ ▷ $\hat{P}_{kv} \in \mathbb{R}^{D_{kv} \times 4C \times T \times wnd}$

▷ $M_{pose}, M'_{CLS} \in \mathbb{R}^{4C \times T \times 1}$

$O \leftarrow [\hat{P}_q || \text{expand}(M_{pose}, D_q) || \text{expand}(M'_{CLS}, D_q)]$

$\tilde{O} \leftarrow [\hat{P}_{kv} || \text{expand}(M_{pose}, D_{kv}) || \text{expand}(M'_{CLS}, D_{kv})]$

$O \leftarrow O + \text{MSA}(\tilde{O}W_q, \tilde{O}W_k, \tilde{O}W_v)$

$O \leftarrow O + \text{FFN}(\text{LN}(O))$

Algorithm 2 Temporal stride attention

Input: *RGB* tokens Z , *Pose* tokens P , *CLS* modal token M_{CLS} , *RGB* modal token M_{RGB} , *Pose* modal token M_{pose} , window size wnd , query weight W_q , key weight W_k , value weight W_v

$stride \leftarrow \lfloor wnd/2 \rfloor$

$\hat{N}_q \leftarrow \emptyset$ ▷ query set

$\hat{N}_{kv} \leftarrow \emptyset$ ▷ key and value set

$N \leftarrow [Z || P || M_{RGB} || M_{pose} || M_{CLS}]$

$D_n \leftarrow (\frac{H}{8} \times \frac{W}{8} + R + 3)$ ▷ The number of tokens

$i \leftarrow 0$

while $i < (T - wnd)$ **do** ▷ Split N into query set $\hat{N}_q$

$\hat{N} \leftarrow N[:, i : i + wnd, :]$ ▷ $N \in \mathbb{R}^{4C \times T \times D_n}$

$\hat{N}_q \leftarrow \hat{N}_q \cup \hat{N}$ ▷ $\hat{N} \in \mathbb{R}^{4C \times wnd \times D_n}$

$i \leftarrow i + stride$

if $i > (T - wnd)$ **then**

$i \leftarrow i - wnd$ ▷ To assure query covers entire tokens

end if

end while

$i \leftarrow stride$

while $i < (T - wnd)$ **do**

▷ Split N into key and value set $\hat{N}_{kv}$

$\hat{N} \leftarrow N[:, i : i + wnd, :]$ ▷ $N \in \mathbb{R}^{4C \times T \times D_n}$

$\hat{N}_{kv} \leftarrow \hat{N}_{kv} \cup \hat{N}$ ▷ $\hat{N} \in \mathbb{R}^{4C \times wnd \times D_n}$

$i \leftarrow i + stride$

end while

$N \leftarrow \hat{N}_q + \text{MSA}(\hat{N}_q W_q, \hat{N}_{kv} W_k, \hat{N}_{kv} W_v)$

$N \leftarrow N + \text{FFN}(\text{LN}(N))$

2. Detailed Description of 3D Deformable Attention

In this study, we proposed the 3D deformable attention to adaptively capture not only long-term temporal relations but also spatial relations simultaneously. Inspired by Xia *et al.*, we rebuilt a deformable attention transformer (DAT) applicable with various video tasks including action recognition. Our proposed method finds discriminative tokens across 3D space while the DAT leverages only 2D space tokens. Details are described in Algorithm 3.

3. Model Efficiency

We compare the proposed method with existing transformer models in terms of the number of parameters and computational complexity as shown in Table 3. When we

$$\begin{aligned} & \text{TemporalStrideAttention}(\hat{N}_q W_q, \hat{N}_{kv} W_k, \hat{N}_{kv} W_v) \\ &= \sum_w \sum_n \text{softmax}\left(\frac{\hat{N}_q W_q^{w,n} \hat{N}_{kv} W_k^{w,n}}{\sqrt{d_h}}\right) \hat{N}_{kv} W_v^{w,n} \end{aligned} \quad (6)$$

where $\hat{N}_q$ and $\hat{N}_{kv}$ from Algorithm 2. In this case, because wnd is always less than T , the overall time complexity of temporal stride attention is smaller than full attention.



❖ Tip 16: Checklist again

- Novelty
 - Is it new to people in the community?
- Significant contributions to the community
 - Is it worth to be reported to people in the community?
- Make sure your paper is non-rejectable (above the bar with some error margin)
 - Sometimes bad papers with no obvious reason to be rejected are accepted
 - That's why you see many papers are accepted

Such luck is not yours.



❖ Tip 17: Code open

- Recently, many top conferences are requiring open source code starting from the Review process.

ICML 2024 Author Instructions

Paper Submissions

Authors are encouraged to submit code to foster reproducibility.

Reproducibility of results and easy availability of code will be taken into account in the decision-making process.

Semantic Scene Graph Generation Based on an Edge Dual Scene Graph and Message Passing Neural Network

Anonymous ECCV 2024 Submission

Paper ID #4020

Abstract. Along with generative AI, interest in scene graph generation (SGG), which comprehensively captures the relationships and interactions between objects in an image and creates a structured graph-based representation, has significantly increased in recent years. However, relying on object-centric and dichotomous relationships, existing SGG methods have a limited ability to accurately predict detailed relationships. To solve these problems, a new approach to the modeling multi-object relationships, called edge dual scene graph generation (EdgeSGG), is proposed herein. EdgeSGG is based on an edge dual scene graph and object-relation centric message passing neural network (OR-MPNN), which can capture rich contextual interactions between unconstrained objects. To facilitate the learning of edge dual scene graphs with a symmetric graph structure, the proposed OR-MPNN learns both object- and relation-centric features for more accurately predicting relation-aware contexts and allows fine-grained relational updates between objects. A comparative experiment with state-of-the-art (SoTA) methods was conducted using two public datasets for SGG operations and six metrics for three subtasks. Compared with SoTA approaches, the proposed model exhibited substantial performance improvements across all SGG subtasks. Furthermore, experiment on imbalanced class distributions revealed that incorporating the relationships between objects effectively mitigates existing long-tail problems. Our code is available at <https://anonymous.4open.science/r/EdgeSGG-pytorch-80DE>



❖ Our goals are...

- **For the reviewers:** clarify doubts, answer questions, correct misunderstandings, push back on mischaracterizations, and make a good-faith effort to incorporate feedback and *improve your work*.
- **The AC:** convince them that you have made a good-faith effort, present a representative summary of the reviews, help them understand if the reviewer concerns were addressed, call out bad-faith reviewing, and ultimately, *help them make a decision*.



❖ Process

- 1. Itemize reviewer comments
 - We use a handy spreadsheet to organize individual comments/questions/concerns raised by each reviewer.
 - Laying everything out in the same place helps identify common concerns and keeps us from missing anything accidentally.
 - Do this **ASAP** to identify any necessary new experiments (if the venue allows it) or analysis early.



❖ Process

- 2. Brain dump possible responses
 - The sheet has a column for each author to reply to each reviewer comment.
 - Dump your thoughts here in rough text without regard for style or length.
 - Being convincing and concise is a subtractive process (for next step).



❖ Process

- 3. Write a draft rebuttal
 - Turn your consensus in the sheet into concrete responses in a rebuttal draft.
 - Write concisely but don't worry about space.
 - Cover every point and trim / prioritize them later.



❖ A list of tips and principles for writing rebuttals

■ Tips. 1. Start positive

- Provide a summary of the reviews, highlighting positive things that reviewers said about your work.
- Rebuttals focus mostly on responding to perceived negatives, don't let ACs forget about the strengths along the way.

We appreciate the constructive feedback from the reviewers (**R1**, **R2**, **R3**). The reviewers acknowledged effectiveness (**R2**, **R3**) and novelty (**R1**, **R2**, **R3**). Below, we address all the questions raised by the reviewers. We will revise to include all these answers in the final version.



❖ A list of tips and principles for writing rebuttals

■ Tips. 2. Order matters

- Start with the **biggest concerns** that you have good answers for and work your way to less clear-cut responses and minor points.



❖ A list of tips and principles for writing rebuttals

■ Tips. 3. Respond directly.

- Quote the core of the reviewer's question or concern concisely and completely. Then before saying anything else, **respond to it directly**. And then give details, describe context, or explain your position.

[R3] Given the use of Conceptual Captions (CC), are the comparisons to baselines fair? We believe these comparisons are fair. We agree that CC is a large, additional data source; however, being able to leverage this additional data for a diverse range of vision and language tasks is precisely our contribution! Existing approaches to vision and language tasks are simply not designed to do so – for instance, it is unclear how to train a standard VQA model like BAN with CC captioning data. Arguing from analogy, the widespread transfer of deep models pretrained on ImageNet also leveraged more data during pretraining; however, we do not find it unfair to pre-deep learning approaches that were not equipped to leverage that data. Finally, note that unlike ImageNet, CC is webly supervised, and did not involve expensive human annotation. We acknowledge that in caption-based image retrieval, CC data could have been used to pretrain existing work for a more direct architectural comparison – we will address.



❖ A list of tips and principles for writing rebuttals

■ Tips. 4. Be conversational

- Make sure your response is **conversational**.....
- It makes it easier for ACs to follow, and the responses are less likely to be perceived as being aggressive.



❖ A list of tips and principles for writing rebuttals

- Tips. 5. Respond to the intent of the questions.
 - If a reviewer points out something with malicious intent, identify the intent and strongly refute it. Otherwise, AC may misunderstand that the criticism is justified.

@R1 – Verification through human-study not scalable:

Our evaluation is not entirely based on human studies. In fact, most of our evaluation is quantitative – see Section 5 and 6 where we quantitatively evaluate task performance and grounding, both of which show the effectiveness of HINT without requiring human studies. We conduct human studies just to evaluate whether HINTed models are more trustworthy to humans than base models.



- ❖ **A list of tips and principles for writing rebuttals**
 - **Tips. 6. Don't be afraid of emphasis.**

Rebuttal:

Q1. The proposed Cook conceptually is similar to a recent video scene graph generation approach [1, 2], which also introduces the knowledge matrix to facilitate context understanding. The authors are encouraged to discuss the connection and the difference between these approaches.

A1. Thank you for your valuable comments. Although your recommended papers [1] and [2] are similar in that they consider co-occurrence, **their solutions and contributions are completely different from ours. We summarize the key differences between our method and recommended papers [1][2].**

1. Domain

Both of the papers you mentioned, **[1] and [2]**, are models for Scene Graph, but they **operate in the video domain** rather than the still image domain. Therefore, the way their models work for Scene Graph generation is completely different. **Still images are not sequential data, so inference happens within a single image, while video-based scene graph generation works by referencing information between neighboring frames. Therefore, a direct comparison of the models is not fair.**



❖ A list of tips and principles for writing rebuttals

- **Tips. 7. Get credit for details you already included.**
 - That said, if something a reviewer asked for was already in the paper, say so. Give them line/Table/Figure numbers, and then restate it in the rebuttal.

[R1: baseline that retrieves images from caption]

As described in L341, 746, 772, round 0 is the caption. Thus the round 0 point in Fig 4a is the caption-based image retrieval baseline. At the best-performing round, RL-Full-QA achieves ~3% percentile (~285 absolute rank) improvement over RL-Full-QA round 0.



❖ A list of tips and principles for writing rebuttals

- **Tips. 8. Express common concerns in a unified way..**
 - Save space by responding to multiple reviewers at once if they share related concerns.

@R2,R3 – Why gradient-based explanations? Why not align attention to human importance? Gradient explanations directly link model decisions to input regions (being decision gradients after all) and so aligning these importances ensures the model is basing its decision on human-attended regions. Further, gradient-based explanations are a function of all the parameters of the network (L443-447). In contrast, attention is a bottom-up computation that relies only on the image and the question (L483). Even with appropriate attention, the remaining network layers may still disregard the visual signal in the presence of strong biases in the dataset.



❖ A list of tips and principles for writing rebuttals

■ Tips. 9. Let's use color-code for reviewers.

- Make it as easy as possible for reviewers to spot responses that are relevant to them

@R2,R3 – Why gradient-based explanations? Why not align attention to human importance? Gradient explana-

■ Tips. 10. Stats speak louder than words.

- Rather than argue with ACs, give them data/stats to back your claim up. These can be statistics/analysis of your data or results.



❖ A list of tips and principles for writing rebuttals

- Tips. 11. Don't promise, **do** !.
 - Instead of saying "We will discuss Singh et al. in the paper.", provide a discussion in the rebuttal.
 - And then also add that you will add it to the paper. It makes it significantly easier for ACs to trust that you will make the promised changes.

@R1, dialog-level evaluation: Thanks for the suggestion! Using Recall@5 to define round-level 'success', our best discriminative model MN-QIH-D gets 7.01 rounds out of 10 correct, while generative MN-QIH-G gets 5.37. Further, the mean first-failure-round (under $R@5$) for MN-QIH-D is 3.23, and 2.39 for MN-QIH-G. Fig. 1c and Fig. 1d show plots for all values of k in $R@k$. We will add this analysis.



❖ A list of tips and principles for writing rebuttals

■ Tips. 12. Be transparent.

- Reviewers asked for additional experiments in rebuttal, but the conference won't allow them? Say so.

R1, R3 Can self-supervised pretraining be skipped given the large amount of data in the multi-task setting? This is an exciting experiment that we have not investigated! We began with pretrained ViLBERT in order to start from a near SOTA trunk model. It may be that training under multi-task supervision provides enough information that large-scale self-supervision is not needed. We will try this and report results in the camera ready as we cannot report it here.



❖ A list of tips and principles for writing rebuttals

- Tips. 13. Highlight abusive reviewers.
 - It may be helpful to point out the reviewer's unreasonable or unsubstantiated comments and persuade the AC to refer to the comments of other reviewers who disagree with the abusive review.
 - Separate, confidential comments may be made on AC.

@R1 – *“It is almost common sense that as the % of the dataset with HINT supervision gets higher, the performance will definitely get higher”*: We disagree. It is not obvious that access to additional, even if relevant, information during training will necessarily improve performance **when generalizing to test instances without this information** – how the additional information is used is paramount. In

Unfair Review and Score by Reviewer jfB2: Request for Reconsideration

[Edit](#)

Confidential Comment to AC | Authors (Hyeongjin Kim, Dasom Ahn, Sangwon Kim, Jong Taek Lee, +1 more) | 06 Apr 2024, 15:38

Program Chairs, Senior Area Chairs, Area Chairs, Authors

Comment:

Dear Program and Area Chairs,

We are writing to express our concerns regarding Reviewer jfB2's comments and score. We believe that Reviewer jfB2's primary critique, which resulted in a low score, is based on a misunderstanding of our work.

Reviewer jfB2 asserted that our proposed method is similar to existing papers [1] and [2].



- ❖ **A list of tips and principles for writing rebuttals**
 - Tips. 14. Acknowledge reviewer efforts.

We appreciate the constructive feedback from the reviewers (**R1**, **R2**, **R3**). The reviewers acknowledged effectiveness (**R2**, **R3**) and novelty (**R1**, **R2**, **R3**). Below, we address all the questions raised by the reviewers. We will revise to include all these answers in the final version.



❖ A list of tips and principles for writing rebuttals

Forty-first International Conference on Machine Learning

Your Submissions		Author Tasks	
#	Paper Summary	Reviews	Decision
435	Scene Graph Generation Strategy with Co-occurrence Knowledge and Learnable Term Frequency Download PDF Hyeongjin Kim , Sangwon Kim , Dasom Ahn , Jong Taek Lee , Byoung Chul Ko ICML 2024 Conference Submission Show details	4 Reviews Submitted Reviewer jfB2: Rating 3 Confidence: 5 Read Review Reviewer MGU2: Rating 4 Confidence: 4 Read Review Reviewer xzQn: Rating 4 Confidence: 5 Read Review Reviewer UMhn: Rating 6 Confidence: 4 Read Review Average Rating: 4.25 (Min: 3, Max: 7) Average Confidence: 4.5 (Min: 4, Max: 5)	ICML 2024 Conference Submission No Recommendation Complaint to AC about reviewer's insincere and malicious review Add: Withdrawal Confidential Comment to AC Unfair Review and Score by Reviewer jfB2: Request for Reconsideration Confidential Comment to AC / Authors (Hyeongjin Kim, Dasom Ahn, Sangwon Kim, Jong Taek Lee, +1 more) 06 Apr 2024, 15:38 Program Chairs, Senior Area Chairs, Area Chairs, Authors Comment: Dear Program and Area Chairs, We are writing to express our concerns regarding Reviewer jfB2's comments and score. We believe that Reviewer jfB2's primary critique, which resulted in a low score, is based on a misunderstanding of our work.



❖ Rebuttal forms

■ CMT

- <https://cmt3.research.microsoft.com/>
- ICML22, AAAI22~24, ICCV23, WACV23~24. ECCV24
- Only one page

View Author Feedback

Paper ID	2463
Paper Title	Concept Graph Embedding Models for Enhanced Interpretability and Accuracy
Track Name	AAAI2024

AUTHOR FEEDBACK QUESTIONS

1. Please upload a maximum 1 page pdf of the author response. The pdf should be formatted using the same template as the submission. Please note that any violation (exceeding the page limit and/or changing the margins) can result in the rejection of the paper.

Agreement accepted

AUTHOR FEEDBACK FILES

[AAAI24 - CGEM_rebuttal.pdf](#) (559 Kb, 2023. 11. 8. 오후 12:58:19)



❖ Rebuttal forms

■ CMT

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Cross-Modal Learning with 3D Deformable Attention for Action Recognition

We appreciate the constructive feedback from the reviewers (R1,R2,R3). The reviewers acknowledged effectiveness (R2,R3) and novelty (R1,R2,R3). Below, we address all the questions raised by the reviewers. We will revise to include all these answers in the final version.

Recent Transformers for Action (R1,R3). We conducted additional analysis and discussed recent studies in action recognition based on transformers. DVT¹ incorporates space-time deformable attention in ViTs to address cost and fixed view limitations. However, the deformable offsets at each layer are based on RGB-level motion cues, which limit diverse representation. Video transformers like MVD² and VideoMAE V2³ enable powerful representation learning through the restoration of masked videos. LART⁴, a tube-based action recognition model, demonstrates outstanding performance but involves complex tasks. *We will add the discussion at the first paragraph of section 2.*

Cross-modal Tokens (R1,R2,R3). In our approach, the cross-modal tokens are concatenated in a channel-wise manner and then passed through the head composed of LN and FC layers, as shown in the equation: $y = FC(LN(M)); M = [M_{RGB} || M_{pose} || M_{CLS}]$. The final class prediction y are minimized using LabelSmoothingCrossEntropy. *We will add the details at line 574.* To analyze the impact of the cross-modal tokens, we conducted ablation studies as shown in Table 5(New). In the ‘No tokens’ scenario, averaging of the last RGB and pose tokens led to a performance drop to 92.2%. Using a single token for all modalities resulted in a slight decrease to 97.0%. However, our approach achieved the best performance, highlighting the effectiveness of learning multiple modalities in a single transformer design. *We will add the results in section 4.2.*

3D Deformable Attention (R1,R2,R3). The DAT⁵ struggles with diverse modalities commonly used in action recognition, and adding separate modal networks and frame-by-frame processing increases complexity. Based on the feedback from reviewers, we analyzed the contribution of our approach and demonstrated its relationship with full attention as follows. 1) Relationship with full attention - We demonstrated our approach exhibits strong performance to the extent of replacing full attention, as shown in Table 4(Mod) ②, ④ and ⑥. 2) Contribution of 3D deformable attention in the RGB modality - In Table 4(Mod) ③ and ⑤, we validated the effectiveness of our method.

Table 4(Mod). Ablation study with attention blocks on PennAction.

	Modality	3D def.	Joint	Temp.	Top-1
①	R+P		✓	✓	94.8
②	R	Full			95.8
③	R	✓			97.9
④	R	Full	✓		96.9
⑤	R	✓		✓	98.2
⑥	R+P	Full	✓	✓	98.2
⑦	R+P	✓	✓	✓	98.5
Ours	R+P	✓	✓	✓	99.7

Table 5(New). Comparison of the Top-1 accuracies for different modal token configurations on PennAction.

Modal Tokens	Top-1 (%)
No tokens	92.2
Single token	97.0
Cross-modal tokens	99.7

Table 6(New). Comparison of complexity with other multi-modal-based human action recognition methods.

Method	Clip size	GFLOPs	Params (M)	Latency (ms)
TSMF	1 × 225 ²	2	14	32
MMNet	1 × 225 ²	2	14	33
MMTM	8 × 224 ²	63	56	38
STAR	16 × 224 ²	53	13	194
Ours	12 × 224 ²	35	18	41

To convey the discussions more clearly, we will add the results of 1) and a modality column in Table 4(Mod).

Model Complexity (R3). We compared our approach with other multi-modal-based human action recognition studies in terms of complexity, as shown in Table 6(New). Most methods lacked reproducible code, making it difficult to compare them. We could only conduct experiments on four selected methods. MMTM and STAR, which handle temporal features across the entire model, exhibited higher complexity and a significant latency of 194ms. In contrast, our proposed model, despite being transformer-based, showed low computational requirements and latency. *We will add the results in section 4.2.*

Position Embedding on pose (R3). pose tokens always possess unique characteristics of individual joints, which means each token is consistently associated with a specific position. Therefore, pose tokens maintain a fixed permutation, and thus we did not apply position embedding to pose tokens. The pose tokens are provided to the joint stride attention, where they are flattened into $T \times wnd$ within each sliding window (wnd). *We will add the details at line 485.*

Misc. (R3). 1) Line: 313-314 - ①intermediate feature maps - refers to the second and fourth feature maps of the backbone, ②sequential tasks - indicates that the 3D conv backbone processes sequential data. *We will add the details at line 314.* 2) Improving readability - *We will ensure readability by following procedures:* ①Reordering the description sequence according to Fig. 2, and ②Ensuring consistency in terminology between the figure and the main text. Through these steps, we aim to convey the research content clearly and facilitate the understanding of the readers.

¹J. Wang et al. Deformable Video Transformer. In CVPR 2022.

²R. Wang et al. Masked Video Distillation. In CVPR 2023.

³L. Wang et al. VideoMAE V2. In CVPR 2023.

⁴J. Rajasegaran et al. On the Benefits of 3D Pose and Tracking for Human Action Recognition. In CVPR 2023.

⁵Z. Xia et al. Vision Transformers with Deformable Attention. In CVPR 2022.



❖ Rebuttal forms

- OpenReview
 - <https://openreview.net/>
 - ICML23~24, NeurIPS22~24, ICLR13~24
 - No page limit, but no images

How we write rebuttals



❖ Rebuttal forms

■ OpenReview

Rebuttal by Authors

Rebuttal

Authors

Hyeongjin Kim, Dasom Ahn, Sangwon Kim, Jong Taek Lee, +1 more

28 Mar 2024, 09:34 (modified: 29 Mar 2024, 21:27)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions

Rebuttal:

Thank you for careful comments in the following parts, we address your concerns:

Q1. Related work on object co-occurrence is missing.

A1: As you suggested, we will add the following related work on object co-occurrence to the revised paper. The content to be added after Line 136 is as follows:

2.3 Label Correlation

After Line 136: "Several approaches have been attempted in various fields to incorporate the correlation between labels into learning. For multi-label classification, SSGL [1] utilized a graph structure defined by co-occurrence information between labels to improve label classification performance. SALGL [2] achieved higher performance multi-label classification than previous studies by simultaneously utilizing co-occurrence knowledge and scene-aware knowledge between labels. In the SGG field, BA-SGG [3] enabled more sophisticated scene graph generation by utilizing the co-occurrence information of "predicates" between objects. However, it has the limitation of not considering the co-occurrence knowledge between objects."

Additional References

[1] Tianshui Chen, Muxin Xu, Xiaolu Hui, Hefeng Wu, and Liang Lin. Learning semantic-specific graph representation for multi-label image recognition. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2019.

[2] Zhu, Xuelin, et al. Scene-aware label graph learning for multi-label image classification. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). 2023.

[3] Guo, Y., Gao, L., Wang, X., Hu, Y., Xu, X., Lu, X., Shen, H.T., & Song, J. From General to Specific: Informative Scene Graph Generation via Balance Adjustment. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2021

Rebuttal by Authors

Rebuttal

Authors

Hyeongjin Kim, Dasom Ahn, Sangwon Kim, Jong Taek Lee, +1 more

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Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions

Rebuttal:

Q2. Zero-shot evaluation missing

A2: We completely agree with your concern that when using prior knowledge, performance may decrease if the distribution of the applied data does not match. To address this concern, we performed zero-shot experiments as suggested and presented the results in Table A. The experiments were conducted on the VG dataset. The experimental results show that the proposed model competitive performance against the SoTA models even in zero-shot settings. Even though our method was not designed for zero-shot purposes, it is meaningful that these results were obtained in a zero-shot experiment. This indicates that the extracted Cook prior knowledge well captures the co-occurrence probabilities in the real world. Since the probability of simultaneous occurrences that can be observed in the real world is infinite, it cannot reflect all situations in the real world. However, various situations not reflected in Cook can be alleviated through a sampling strategy using additional data.

Table A. Comparison on zero-shot evaluation with SoTA methods using VG dataset

Methods	PredCls zs-R@50 / 100	SGCls zs-R@50 / 100	SGGen zs-R@50 / 100
VCTree [1]	3.3 / 5.5	1.2 / 2.1	0.3 / 0.7
MORIFS [2]	3.2 / 5.4	0.7 / 1.1	0.1 / 0.1
FC-SGG [3]	7.9 / 10.1	1.7 / 2.1	0.9 / 1.4
IS-GCT [4]	8.3 / -	2.6 / -	1.0 / -
PE-Net [5]	17.2 / 20.9	5.4 / 6.5	2.3 / 3.6
Cook + TF- LIDF	12.7 / 14.4	6.7 / 7.1	1.3 / 1.4

Additional experiments related to Zero-shot oriented SGG SoTA methods

[1] Hengyue Liu, Ning Yan, Masood Mortazavi, and Bir Bhanu. Fully convolutional scene graph generation. CVPR, pp. 11546–11556, June 2021

[2] Rowan Zellers, Mark Yatskar, Sam Thomson, and Yejin Choi. Neural motifs: Scene graph parsing with global context.CVPR, pp. 5831–5840, 2018

[3] Kaihua Tang, Hanwang Zhang, Baoyuan Wu, Wenhan Luo, and Wei Liu. Learning to compose dynamic tree structures for visual contexts. CVPR, June 2019.

[4] Kundu, S. and Aakur, S. N. Is-ggt: Iterative scene graph generation with generative transformers. CVPR, pp. 6292–6301, 2023.

[5] Zheng, C., Lyu, X., Gao, L., Dai, B., and Song, J. Prototypebased embedding network for scene graph generation. CVPR, pp. 22783–22792, 2023.

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❖ **Q1. it is challenging to come up with an entirely new model/design in a short time.**

- Modify the existing model/design by adding a new component ? how much innovation is required to be accepted in the latter case.
- A:
 - Among the many papers in the Top Conferences, there is nothing completely innovative or new (except for exceptional cases such as CNN, LSTM, Transformer, GAN, Diffusion, etc.).
 - Most methods are used to improve previous methods or combine different algorithms. Recently, there are many papers that combine generative AI or large language models in their research.



❖ Q1. it is challenging to come up with an entirely new model/design in a short time.

■ A:

- So now Focus on, how do we creatively combine existing models (unlike before)? And how much will the performance improve as a result?

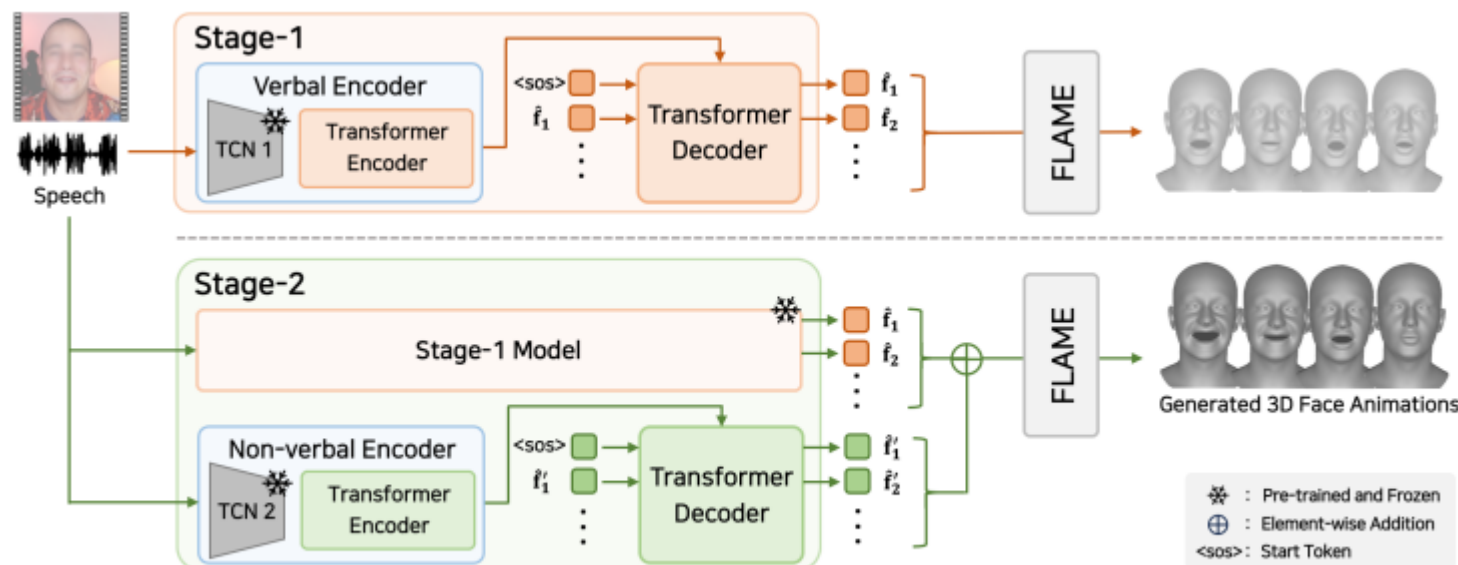


Figure 3. **LaughTalk architecture.** The stage-1 model extracts verbal features from the input speech and generates facial motion



❖ **Q2. most top-tier conferences do not like application-oriented or theoretical papers, is there any suitable conference to publish such kinds of papers in AI?**

■ **A:**

- The preference varies depending on the type of top conferences. For ICML, AAAI, ICLR, and NeurIPS, they prefer fundamental algorithms over applications.
- In contrast, CVPR, ECCV, ICCV, WACV, etc., favor trendy applied research. However, it's crucial to remember that both types of conferences demand novelty.
- Simply combining models doesn't make it innovative.



- ❖ Q3. conferences have strict page limits and detailed results (graphs) cannot be adequately discussed, is there any suggestions in this regard for naive researchers?

4020	Semantic Scene Graph Generation Based on an Edge Dual Scene Graph and Message Passing Neural Network Show abstract	Submission files: ⬇ main.pdf Supplementary Files: ⬇ EdgeSGG ECCV2024 Appendix.pdf
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- ❖ **Q4. what are some latest trends to be accepted in the top-tier conferences? It would be better to share some latest topics/trends for the conference papers?**
 - **A:** Refer the front pages



❖ **Q5. what are the differences in writing style for journals and conferences in the near future? Some useful tips in this regard can be shared?**

- **A:** There is not much difference between journals and conferences.
 - Conferences tend to emphasize algorithms and experiments rather than Intro. related works, conclusion bz. limitation of pages. Journals still require deail coverage of Intro. & related works.



- ❖ **Q5. what are the differences in writing style for journals and conferences in the near future? Some useful tips in this regard can be shared?**
- **A:** In the future, it is expected that journals will also transition to open-access formats. Especially in rapidly advancing AI research, traditional top journals (IEEE Trans., ACM Trans., Elsevier, Springer) are also expected to transition to open-access formats.
 - It is anticipated that academia and industry will increasingly recognize the value of top conferences over journals.



- ❖ **Q6. in journal papers, when researchers compare their results with conference papers, it is regarded as less credible and reviewers usually regard it as not SOTA.**
 - **A:** I think the opposite is true. It is also common for journals to request comparison with SoTA results from top conferences as a comparative study.
 - However, if the conference paper is not from a top conference, reviewers may not trust the results.



❖ **Q6. Can we compare our results with conference papers only, journal papers only, or both? What is the best (or common) practice in this regard??**

- **A:** As we all know, top journals (IEEE Trans.) take at least 1 year and 6 months from review to (online) publication.
- Therefore, the journal paper we are looking at is not SoTA. A good way is to compare the results of the most recent top conference papers while also comparing the most recent top journals.
- However, the priority is still on the top conference papers.



My first drafts are so-so, but I think I re-write pretty well. Good writing is re-writing.

This means you need to start writing the paper early!



- • Bill Freeman's how to write papers:
 - https://faculty.cc.gatech.edu/~parikh/citizenofcvpr/static/slides/freeman_how_to_write_papers.pdf
- Jitendra Malik's how to write good papers:
 - https://faculty.cc.gatech.edu/~parikh/citizenofcvpr/static/slides/malik_write_good_paper.pdf
- Jia-Bin's awesome tips:
 - <https://github.com/jbhuang0604/awesome-tips>
- Devi's rebuttal guide: <https://deviparikh.medium.com/how-we-writerebuttals-dc84742fece1>
- PhD comics: <https://phdcomics.com/>
- How we write rebuttals, Devi Parikh



Thanks & Good Luck